SMITHIAN FOLD THEORY - REVERSIBLE AND QUANTUM COMPUTATION PAPER 001

The Quantum Fold Machine

An Exact, Parameter-Free and Machine-Closed Derivation of Reversible and Quantum Computation from Smithian Fold Theory

Ernos Labs

Open Source Science Platform and Knowledge Tree

Maria Smith

Independent researcher and founder, Ernos Labs

Maria.Smith.Sftoe@gmail.com

Published same-paper roadmap successor version 1.3.0

27 July 2026

DOI: 10.5281/zenodo.21627748

PUBLISHED OPEN-ACCESS VERSION

Abstract

This version 1.3 paper preserves the complete registered Reversible and Quantum Computation inventory of the third clean-room reconstruction of Smithian Fold Theory. From the admitted Foundation, Mathematics, Information Science and Classical Computation receipts it derives a complete reversible model; Fold quantum information units and composition; superposition-equivalent support; phase and interference; entanglement; measurement; gates and circuits; quantum universality, algorithms and complexity; communication; coding, error correction and fault tolerance; simulation; verification; learning; full operational classical-quantum correspondence; and the limits of quantum computation. It imports no complex amplitude, Hilbert-space axiom, stochastic collapse postulate, fitted parameter or physical benchmark as a premise.

The frozen branch contains 22 dependency-ordered claims and 5,632 generated candidate structures. Every candidate is decided, each grammar has exactly one survivor, and every claim passes minimality, named-shape uniqueness, depth-independent closure, four adverse controls, cryptographic sealing and implementation-distinct recomputation.

Keywords: Smithian Fold Theory; reversible computation; quantum information; superposition; phase; interference; entanglement; measurement; quantum circuits; error correction; fault tolerance; quantum algorithms; quantum complexity; open computational science

1. Central scientific claim and exact boundary

> *Within the current SFT V3 Reversible and Quantum Computation inventory, all 22 registered laws and all 29 Quantum Computation-owned V1/V2 reconstruction obligations have depth-independent, model-admitted and independently replicated receipts, and no registered quantum-computation obligation remains unclassified or frontier.*

Closure is formal and structural. It includes the unbounded positive-finite `2t+1` fault-order theorem, but does not convert that theorem into a measured stochastic hardware threshold. Physical realization, device fidelity, error distributions, natural quantum probabilities and quantum physics require the sealed empirical Physics branch.

2. Headline results at a glance

Result	Exact closed statement	Executed boundary
One quantum information unit	One Fold distinction supplies the complete two-held-label information unit without a complex amplitude premise.	Exact finite word support, observation classes and retained phase-label carrier.
Superposition and interference	Superposition-equivalent structure is complete generated branch support; phase is period-held action; interference is exact predecessor merging with phase provenance.	Complete branch-to-image relations and inverse records; no stochastic collapse postulate.
Entanglement	A joint state is entangling exactly when its complete pair-cell support is nonfactorable relative to its marginals.	Product and correlated supports, component identities and exact marginal reconstruction.
Measurement	Observation retains one exact class and a complete source/phase/image record, making support reduction reconstructible at the declared record boundary.	Every pre-observation branch and closed distinction is recorded.
Unbounded finite fault order	For every supplied positive finite `t`, the unique first correcting width is `2t+1`, redundancy is `2t`, every predecessor width fails, and the successor adds two labels.	Every mask at `t=1,2,3`; all predecessor widths and successors through `t=14`; all 128 depth-seven words at `t=14`.
Quantum computational limits	Every admitted quantum circuit remains a finite generated description, so self-reference, halting and undeclared-oracle boundaries transfer.	Branchwise operational correspondence with the classical reversible submodel.

Historical quantum formalisms and names enter after the Fold carriers, transformations and observations are sealed. They are correspondence tests and cannot select the law, code width, survivor or claimed resource.

Public scientific mission and admission boundary

Ernos Labs is an open-source science movement, verification platform and public tree of knowledge founded by Maria Smith. Its purpose is not to replace one authority with another. It is to make scientific authority narrow, inspectable and revocable: a claim is admitted only through its complete derivation chain, generated alternatives, eliminations, unique survivor, adverse controls, measurement custody where applicable and unchanged-engine receipt. Open criticism is unrestricted and necessary; scientific admission is the separate machine-checked act of satisfying that public standard.

Maria Smith developed Smithian Fold Theory outside formal academic education, institutional research employment and conventional grant funding. That fact is not offered as evidence for a theorem; the derivations and observations carry the entire scientific burden. It is evidence about access. A credential-first system loses more than individual opportunity: it loses unknown questions, methods and discoveries from minds that capital and status never authorize. This work therefore treats Maria Smith's authorship neither as exceptionalism nor as a reason for dismissal, but as an indictment of every scientific contribution lost when financial gatekeeping is presented as rigor.

The institutional argument is empirical, not a claim that every institution or funded researcher acts in bad faith. Published studies document sponsor-linked differences in outcomes and conclusions, commercial influence over research agendas, limits and sensitivities in grant review, underpublication of null results, and inequalities created by paywalls and article-processing charges. Those findings establish that funding, prestige, publication and consensus are selection systems with incentives and failure modes. They cannot substitute for a public proof and evidence chain. Expertise, measurement and adversarial review remain indispensable; institutional permission does not select a fundamental law.

For Quantum Computation, inaccessible hardware summaries, proprietary simulation, fitted amplitude tables and authority labels cannot admit a quantum law. Every support branch, phase action, merge, joint cell, observation record, correction mask and resource must remain inspectable. Hardware measurements may test a separately sealed physical claim, but cannot flow backward to select the formal quantum-computation structure.

Independent replications, lawful extensions, corrections and attempted invalidations are invited. Credentials cannot rescue a failed gate and lack of credentials cannot prevent a reproducible result from being evaluated. Contact Maria.Smith.Sftoe@gmail.com, submit through https://discord.gg/ucwGryVxGr, or inspect the public project at https://github.com/MettaMazza.

Evidence-boundary inventory identity:
`sha256:d421a26e77e356ebb227bbcab026dfe6aaedcd3019dc55d7e70adf2503cfd542`.

3. Fold-native quantum constitution

One Fold distinction supplies the native information unit. Complete held-label words across generated positions supply branch support. A branch carries an exact held phase label from a generated finite cycle. Joint states are complete pair-cell words retaining component identity. A transformation is a total branch-word bijection plus a period-held phase action. None of these definitions requires a real or complex amplitude, irrational normalization, imaginary proof value or ungenerated continuum.

4. Superposition, interference and entanglement

Superposition-equivalent means complete simultaneous support of every generated branch before an observation class is retained; it does not install an ontic stochastic mixture. Phase is a reversible action on held cycle labels. Interference is a complete branch-to-image relation whose predecessor fibres preserve every merging branch and its phase. Joint support is factorable exactly when it equals the Cartesian product of its marginals; a strict correlated subset is nonfactorable and supplies the entangling class.

5. Measurement and reversibility

Measurement is a total observation relation over complete branch support. The selected observation class becomes the retained state, while the measurement record keeps every pre-observation branch, phase and image label. Support reduction is therefore exact and reconstructible at the record boundary. A gate is reversible only when its branch map is bijective and its phase action has an exact inverse; any predecessor merge requires a retained predecessor label.

6. Coding, multi-error correction and unbounded finite fault order

A generated positive fault-order trace of length t forces repetition width 2t+1: no more than t labels change and at least t+1 retain the source, forcing the strict majority. Every width through 2t has a constructive counterexample formed by changing ceil(w/2) positions. The executable census exhausts every mask through one error at width 3, two errors at width 5 and three errors at width 7. It executes every predecessor-width and successor certificate through t=14 and round-trips all 128 depth-seven source words at t=14 before exact circuit evaluation. The successor t to t+1 adds exactly two labels and constructively defeats every fixed positive finite ceiling. No completed infinite-width object, measured error rate or physical hardware threshold is asserted.

7. Classical-quantum correspondence and limits

Classical computation embeds as the single-held-branch, phase-insensitive reversible submodel. Quantum execution extends that carrier with complete branch support, phase-sensitive predecessor merging and nonfactorable joint support. Observation decodes exact classical records. Because every quantum circuit remains a generated finite description executed by an admitted universal process, self-reference and halting boundaries transfer: quantum computation does not acquire an undeclared oracle or hypercomputational escape.

8. Complete execution census

Order	Claim	Candidates	Closure
1	`SFT-QUANTUM-REVERSIBLE-MODEL-001`	256	depth-independent
2	`SFT-QUANTUM-INFORMATION-UNIT-001`	256	depth-independent
3	`SFT-QUANTUM-STATE-COMPOSITION-001`	256	depth-independent
4	`SFT-QUANTUM-SUPERPOSITION-001`	256	depth-independent
5	`SFT-QUANTUM-PHASE-INTERFERENCE-001`	256	depth-independent
6	`SFT-QUANTUM-ENTANGLEMENT-001`	256	depth-independent
7	`SFT-QUANTUM-MEASUREMENT-001`	256	depth-independent
8	`SFT-QUANTUM-GATE-001`	256	depth-independent
9	`SFT-QUANTUM-CIRCUIT-001`	256	depth-independent
10	`SFT-QUANTUM-UNIVERSALITY-001`	256	depth-independent
11	`SFT-QUANTUM-ALGORITHMS-001`	256	depth-independent
12	`SFT-QUANTUM-COMPLEXITY-001`	256	depth-independent
13	`SFT-QUANTUM-COMMUNICATION-001`	256	depth-independent
14	`SFT-QUANTUM-CODING-001`	256	depth-independent
15	`SFT-QUANTUM-ERROR-CORRECTION-001`	256	depth-independent
16	`SFT-QUANTUM-FAULT-TOLERANCE-001`	256	depth-independent

Order	Claim	Candidates	Closure
17	`SFT-QUANTUM-SIMULATION-001`	256	depth-independent
18	`SFT-QUANTUM-VERIFICATION-001`	256	depth-independent
19	`SFT-QUANTUM-LEARNING-001`	256	depth-independent
20	`SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`	256	depth-independent
21	`SFT-QUANTUM-LIMITS-001`	256	depth-independent
22	`SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002`	256	depth-independent

The branch total is **5,632 candidates**, **22 survivors**, **88 adverse controls** and **22 independent validations**.

9. Derivation 1: Complete Fold reversible-computation model

Claim: `SFT-QUANTUM-REVERSIBLE-MODEL-001`

9.1 Scientific necessity and exact theorem

Complete Fold reversible-computation model must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced reversible computation kernel retains canonical configurations; bijective transition rows; retained held labels; exact inverse traces; universal reversible interpretation; complete resource ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

9.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: canonical configurations; bijective transition rows; retained held labels; exact inverse traces; universal reversible interpretation; complete resource ledger.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for reversible computation.

> *All finite reversible computation structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. This theorem is formal and does not assign thermodynamic energy without a later physical measurement law.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

9.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced reversible computation kernel retains canonical configurations; bijective transition rows; retained held labels; exact inverse traces; universal reversible interpretation; complete resource ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

9.4 Operational laws and consequences

- Support law: canonical configurations; bijective transition rows; retained held labels; exact inverse traces; universal reversible interpretation; complete resource ledger.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

9.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact reversible computation record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

9.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: `PASS`.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: `PASS`.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: `PASS`.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: `PASS`.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: `PASS`.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: `PASS`.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: `PASS`.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS

Control	Expected	Observed	Result
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; This theorem is formal and does not assign thermodynamic energy without a later physical measurement law.	`PASS`

9.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: reversible computation, Bennett machine.

This theorem is formal and does not assign thermodynamic energy without a later physical measurement law.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- This theorem is formal and does not assign thermodynamic energy without a later physical measurement law.

9.8 Exact evidence identities

- Source manifest: `sha256:c5af094fe52290461356bf39e9d76f13ba2b6202e285508bc50ef96cd4b8e86f`
- Complete-census certificate: `sha256:dd9994b76e30d066ca995523aa7528f47f985b73200c8e677750fa7f652c5033`
- Derivation seal: `sha256:37e9625bca1b99f14969057b963782c6b1cb62efb98ec317178cb32901e2f48c`
- Independent implementation: `sha256:088141be2c397e4c3f7e640dbc6ba3850bc936eb3433537b058e95a25e98e2df`
- Independent certificate: `sha256:33e527c37a50844de7b1b6228f92b22f4eb6a5279fe576a97b952c28885c5ed1`
- External validation: `sha256:ded9f371a7b58164c55628cc9684d6944f8083be860caea6cb43f4f0a629bb60`
- Model-admitted engine receipt: `sha256:068b389a38d47c06118600324bc263c951e6e19c6f591f257c601f6125525298`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-REVERSIBLE-MODEL-001-068b389a38d47c06.json`

10. Derivation 2: Fold quantum information-unit law

Claim: `SFT-QUANTUM-INFORMATION-UNIT-001`

10.1 Scientific necessity and exact theorem

Fold quantum information-unit law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced quantum information units kernel retains one Fold distinction; two held fibre labels; complete word support; exact observation classes; retained phase-label carrier; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-REVERSIBLE-MODEL-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

10.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: one Fold distinction; two held fibre labels; complete word support; exact observation classes; retained phase-label carrier.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum information units.

> *All finite quantum information units structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. The unit is structural support, not an imported complex vector or conventional qubit.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

10.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced quantum information units kernel retains one Fold distinction; two held fibre labels; complete word support; exact observation classes; retained phase-label carrier; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

10.4 Operational laws and consequences

- Support law: one Fold distinction; two held fibre labels; complete word support; exact observation classes; retained phase-label carrier.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

10.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum information units record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

10.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-INFORMATION-UNIT-001; result: `PASS`.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-INFORMATION-UNIT-001; result: `PASS`.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-INFORMATION-UNIT-001; result: `PASS`.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-INFORMATION-UNIT-001; result: `PASS`.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-INFORMATION-UNIT-001; result: `PASS`.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-INFORMATION-UNIT-001; result: `PASS`.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-INFORMATION-UNIT-001; result: `PASS`.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-INFORMATION-UNIT-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; The unit is structural support, not an imported complex vector or conventional qubit.	PASS

10.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: qubit, quantum information unit.

The unit is structural support, not an imported complex vector or conventional qubit.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- The unit is structural support, not an imported complex vector or conventional qubit.

10.8 Exact evidence identities

- Source manifest: `sha256:a27e2cd1d555c89028a74a2426b74986adf84a83e8fac584985522f90e8b0c83`
- Complete-census certificate: `sha256:76b580cc45da482f4fdbd7ad583257f4d3d7719b8062af69b5ac96098cc2f01c`
- Derivation seal: `sha256:8dd1b7f8a5b144885d8926f1d724e9f1ecc767f71ce3110a1884d03297673cdb`
- Independent implementation: `sha256:3a82e4d5098bfd4d453af17516b932f696f816a15c6bc718253472c762d0ee10`
- Independent certificate: `sha256:c7ca84cd3a8c47cd8a5e345697f2648d3861372f3c6a72c96a1288ce660f8ef5`
- External validation: `sha256:7e138a1d7e2f5ce7784a9eab262e857118f8efce2a96a7ec20b6674541cc60a5`
- Model-admitted engine receipt: `sha256:3cae970ebb8336d464bcbc64e17d0f50049da6f018b95f98b53b04a46873bf72`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-INFORMATION-UNIT-001-3cae970ebb8336d4.json`

11. Derivation 3: Fold quantum state-composition law

Claim: `SFT-QUANTUM-STATE-COMPOSITION-001`

11.1 Scientific necessity and exact theorem

Fold quantum state-composition law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced quantum state composition kernel retains complete branch support; exact pair-cell joint words; component identities; phase-label composition; marginal reconstruction; product and correlated classes; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-INFORMATION-UNIT-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

11.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: complete branch support; exact pair-cell joint words; component identities; phase-label composition; marginal reconstruction; product and correlated classes.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum state composition.

> *All finite quantum state composition structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Only finite generated joint supports are admitted.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

11.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced quantum state composition kernel retains complete branch support; exact pair-cell joint words; component identities; phase-label composition; marginal reconstruction; product and correlated classes; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

11.4 Operational laws and consequences

- Support law: complete branch support; exact pair-cell joint words; component identities; phase-label composition; marginal reconstruction; product and correlated classes.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

11.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum state composition record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

11.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS

Control	Expected	Observed	Result
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Only finite generated joint supports are admitted.	PASS

11.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: tensor product, composite state.

Only finite generated joint supports are admitted.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Only finite generated joint supports are admitted.

11.8 Exact evidence identities

- Source manifest: `sha256:e19a28acfd158a2adc8faa62d1c641983e613fd02abd4f5703fd4df71b5a2e32`
- Complete-census certificate: `sha256:d10f6ac72430d0adbf42711c2e21033c6ef6a3fd14d43e9b6aed60ef5336f9d4`
- Derivation seal: `sha256:07accac25aaa131af96d31bdce98536c3ff39c0c0504a69b146f4a02240441c5`
- Independent implementation: `sha256:45b6bcd57a43ea420e30db2d983e862b2206a61b2483270be469b035b5263ee0`
- Independent certificate: `sha256:47fa45ac826615121aeec5d2ea41548aebd01dbac38c96e8e2ee8b359cb507e4`
- External validation: `sha256:7ac24331062df574d62dbb00a5237cdf8bcd928212a5d0baf95a6cef91b867c1`
- Model-admitted engine receipt: `sha256:a358610587ee2c7c6dfcba8a782e46f89a1668e2c86739eac0e8c88d6af719c2`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-STATE-COMPOSITION-001-a358610587ee2c7c.json`

12. Derivation 4: Superposition-equivalent Fold support law

Claim: `SFT-QUANTUM-SUPERPOSITION-001`

12.1 Scientific necessity and exact theorem

Superposition-equivalent Fold support law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced superposition-equivalent structure kernel retains complete b^k Fold-word support; one record per branch; held phase label per branch; no selected actual branch before observation; exact support provenance; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-STATE-COMPOSITION-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

12.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: complete b^k Fold-word support; one record per branch; held phase label per branch; no selected actual branch before observation; exact support provenance.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for superposition-equivalent structure.

> *All finite superposition-equivalent structure structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. No amplitude magnitude, normalization constant or stochastic ontology is imported.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

12.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced superposition-equivalent structure kernel retains complete b^k Fold-word support; one record per branch; held phase label per branch; no selected actual branch before observation; exact support provenance; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

12.4 Operational laws and consequences

- Support law: complete b^k Fold-word support; one record per branch; held phase label per branch; no selected actual branch before observation; exact support provenance.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

12.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact superposition-equivalent structure record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

12.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-SUPERPOSITION-001; result: `PASS`.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-SUPERPOSITION-001; result: `PASS`.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-SUPERPOSITION-001; result: `PASS`.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-SUPERPOSITION-001; result: `PASS`.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-SUPERPOSITION-001; result: `PASS`.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-SUPERPOSITION-001; result: `PASS`.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-SUPERPOSITION-001; result: `PASS`.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-SUPERPOSITION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No amplitude magnitude, normalization constant or stochastic ontology is imported.	PASS

12.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: superposition, basis support.

No amplitude magnitude, normalization constant or stochastic ontology is imported.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- No amplitude magnitude, normalization constant or stochastic ontology is imported.

12.8 Exact evidence identities

- Source manifest: `sha256:0d1eb45fe6d4e484dfcaa6ef334a3ab671ddd0da9f6f153b901cc8351f7603e3`
- Complete-census certificate: `sha256:95ed3e715d21daabab6325f0d1dbeffdd25e7fdaf0c8c481c2bee0c61bd2e0a7`
- Derivation seal: `sha256:3d8ee27d90c7ddfea4b88ece679df71f39ba1602da046edef08f8e3ed4a8bcdd`
- Independent implementation: `sha256:945ffdb45c2324c21427547d49b7e14fbe31aeaca26deee602a921d72088a911`
- Independent certificate: `sha256:0de2881ba9754d71a5ad9d18c7283eafe2a63e31b131dc545c7234b8b668bc63`
- External validation: `sha256:5fd70c68356f3013863b86feacd8f2a221a4002161cc5be3074f29e0c6b719e1`
- Model-admitted engine receipt: `sha256:54c09712a4a53c16054fc2c8de1e247007b718e1f6f0cd9cd38a14b1a5cbe7d1`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-SUPERPOSITION-001-54c09712a4a53c16.json`

13. Derivation 5: Fold phase and interference operation law

Claim: `SFT-QUANTUM-PHASE-INTERFERENCE-001`

13.1 Scientific necessity and exact theorem

Fold phase and interference operation law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced phase and interference kernel retains period-b held phase cycle; reversible phase action; total branch-to-image relation; complete predecessor fibres; same-image merge ledger; retained phase provenance; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-SUPERPOSITION-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

13.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: period-b held phase cycle; reversible phase action; total branch-to-image relation; complete predecessor fibres; same-image merge ledger; retained phase provenance.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for phase and interference.

> *All finite phase and interference structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Interference is exact predecessor merging with phase labels; no complex*

exponential is assumed.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

13.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

The forced phase and interference kernel retains period-b held phase cycle; reversible phase action; total branch-to-image relation; complete predecessor fibres; same-image merge ledger; retained phase provenance; branch

> *support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

13.4 Operational laws and consequences

- Support law: period-b held phase cycle; reversible phase action; total branch-to-image relation; complete predecessor fibres; same-image merge ledger; retained phase provenance.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

13.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact phase and interference record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

13.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS

Control	Expected	Observed	Result
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Interference is exact predecessor merging with phase labels; no complex exponential is assumed.	PASS

13.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: phase, interference.

Interference is exact predecessor merging with phase labels; no complex exponential is assumed.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Interference is exact predecessor merging with phase labels; no complex exponential is assumed.

13.8 Exact evidence identities

- Source manifest: `sha256:e7c78294725fc5b8365f70ee9be63ba92165aa712a7a686ecd999e74a8db7393`
- Complete-census certificate: `sha256:3125c54031dc65d7e5e4963aa4b031acf83c57e009e731ee188d7806d07b7684`
- Derivation seal: `sha256:da60087f0965652227de2f3c71b7584d22e2f16a2b794c4a3fa4152bc27d186a`
- Independent implementation: `sha256:5b0728c303020c2658658987b7794ae08beb9b00f6e2f8c6b48c1cc5251db79b`
- Independent certificate: `sha256:09d8b5b79e90ad34e20b35f399aafd7bc0c8948366962366431e8411290575a0`
- External validation: `sha256:3d7e92436970f0002b6da43d04afbba1374c525fb6b98c43a99f1ab40c60d44f`
- Model-admitted engine receipt: `sha256:c1c7e3f2c71ea7de6d801c397cff01ae8a005651deddcccb7acd83bddc857f12`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-PHASE-INTERFERENCE-001-c1c7e3f2c71ea7de.json`

14. Derivation 6: Fold entangling-composition law

Claim: `SFT-QUANTUM-ENTANGLEMENT-001`

14.1 Scientific necessity and exact theorem

Fold entangling-composition law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced entanglement kernel retains joint word support; exact marginals; product-support test; nonfactorable pair-cell classes; retained component observations; compositional trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-PHASE-INTERFERENCE-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

14.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: joint word support; exact marginals; product-support test; nonfactorable pair-cell classes; retained component observations; compositional trace.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for entanglement.

> *All finite entanglement structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Entanglement is structural nonfactorability of registered joint support, not a metaphysical postulate.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

14.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced entanglement kernel retains joint word support; exact marginals; product-support test; nonfactorable pair-cell classes; retained component observations; compositional trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

14.4 Operational laws and consequences

- Support law: joint word support; exact marginals; product-support test; nonfactorable pair-cell classes; retained component observations; compositional trace.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

14.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact entanglement record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

14.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-ENTANGLEMENT-001; result: `PASS`.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-ENTANGLEMENT-001; result: `PASS`.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-ENTANGLEMENT-001; result: `PASS`.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-ENTANGLEMENT-001; result: `PASS`.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-ENTANGLEMENT-001; result: `PASS`.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-ENTANGLEMENT-001; result: `PASS`.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-ENTANGLEMENT-001; result: `PASS`.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-ENTANGLEMENT-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Entanglement is structural nonfactorability of registered joint support, not a metaphysical postulate.	PASS

14.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: entanglement, nonseparable state.

Entanglement is structural nonfactorability of registered joint support, not a metaphysical postulate.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Entanglement is structural nonfactorability of registered joint support, not a metaphysical postulate.

14.8 Exact evidence identities

- Source manifest: `sha256:afb46ed17a896275cc6922c9f10115e35b92520aa8f0e38eb4bf67a7399cd1c8`
- Complete-census certificate: `sha256:3dccccf9a2fd85f710d29322e1e2200163c35004bfa3d73c701a6520b7404c6d`
- Derivation seal: `sha256:d7fe096c012e1467e6d8253f0d04d02de4bc66f76b479cca2190386dc4298cec`
- Independent implementation: `sha256:ebd4b53c6a0c5b77e9d0f9fc9f7f031c0d9f1f9a77ba9fd886f9153df0e39966`
- Independent certificate: `sha256:cfeb4e7a31d71a6197bc757243682f2fa66e4ce558a1284d2d35c0fd76a7dc43`
- External validation: `sha256:57031064581dc6356636b1f6f65f980795d963109bd56945287e91e13dfb9657`
- Model-admitted engine receipt: `sha256:6bfa64c6f896c5f7eaa3a55872f31cd487923853353a3069ec7a839afa1d3d0d`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-ENTANGLEMENT-001-6bfa64c6f896c5f7.json`

15. Derivation 7: Fold quantum measurement-semantics law

Claim: `SFT-QUANTUM-MEASUREMENT-001`

15.1 Scientific necessity and exact theorem

Fold quantum measurement-semantics law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced measurement kernel retains total observation relation; retained observation class; closed branch distinctions; complete pre-observation record; selected-class state; exact reconstruction boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-ENTANGLEMENT-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

15.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: total observation relation; retained observation class; closed branch distinctions; complete pre-observation record; selected-class state; exact reconstruction boundary.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for measurement.

> *All finite measurement structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. No stochastic collapse postulate is imported; outcome frequencies require external empirical evidence.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

15.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.

Eliminated candidates	Exact first-failure reason
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced measurement kernel retains total observation relation; retained observation class; closed branch distinctions; complete pre-observation record; selected-class state; exact reconstruction boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

15.4 Operational laws and consequences

- Support law: total observation relation; retained observation class; closed branch distinctions; complete pre-observation record; selected-class state; exact reconstruction boundary.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

15.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact measurement record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

15.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-MEASUREMENT-001; result: `PASS`.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-MEASUREMENT-001; result: `PASS`.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-MEASUREMENT-001; result: `PASS`.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-MEASUREMENT-001; result: `PASS`.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-MEASUREMENT-001; result: `PASS`.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-MEASUREMENT-001; result: `PASS`.

- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-MEASUREMENT-001; result: `PASS`.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-MEASUREMENT-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No stochastic collapse postulate is imported; outcome frequencies require external empirical evidence.	PASS

15.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum measurement, collapse.

No stochastic collapse postulate is imported; outcome frequencies require external empirical evidence.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- No stochastic collapse postulate is imported; outcome frequencies require external empirical evidence.

15.8 Exact evidence identities

- Source manifest: `sha256:13904de9dde6457d62a60404ddea441608fa18f0c92f149958bb94b8333e32cf`
- Complete-census certificate: `sha256:2f837ffe8611aa00de3dde2d72e76ac5b16ec836ee79b879b5d3379af0046e71`
- Derivation seal: `sha256:7d99cd7ce254a64f69ce210b7b3883177227978c792faeeb846094c4856dc9fa`
- Independent implementation: `sha256:09615ec981ffda13c0fd1ea5163e580708612ce2202d5b23d98d1c3b806d66c7`
- Independent certificate: `sha256:59226eca7492c0aebb48b4ab677e9b24e1bcfc38d8438d37385b00ed1ab13d1f`
- External validation: `sha256:0b403b2a032ee01b1b8bbc6ae03c9a26b51f564dd38d11f1bc0d775672d89026`

- Model-admitted engine receipt:
  `sha256:e29a71f7dff97835e28850ee8d352ac80aa9ceabb4399e07105d892175776a06`
- Receipt path:
  `receipts/engine/model_admitted/SFT-QUANTUM-MEASUREMENT-001-e29a71f7dff97835.json`

16. Derivation 8: Fold reversible transformation and gate law

Claim: `SFT-QUANTUM-GATE-001`

16.1 Scientific necessity and exact theorem

Fold reversible transformation and gate law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced quantum transformations and gates kernel retains bijective branch-word map; period-held phase action; complete domain and image support; exact inverse; source-bound gate identity; composition interface; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-MEASUREMENT-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

16.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: bijective branch-word map; period-held phase action; complete domain and image support; exact inverse; source-bound gate identity; composition interface.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum transformations and gates.

> *All finite quantum transformations and gates structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Physical gate fidelity remains an empirical engineering question.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

16.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced quantum transformations and gates kernel retains bijective branch-word map; period-held phase action; complete domain and image support; exact inverse; source-bound gate identity; composition interface; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

16.4 Operational laws and consequences

- Support law: bijective branch-word map; period-held phase action; complete domain and image support; exact inverse; source-bound gate identity; composition interface.

- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

16.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum transformations and gates record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

16.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-GATE-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-GATE-001; result: PASS.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-GATE-001; result: PASS.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-GATE-001; result: PASS.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-GATE-001; result: PASS.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-GATE-001; result: PASS.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-GATE-001; result: PASS.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-GATE-001; result: PASS.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Physical gate fidelity remains an empirical engineering question.	PASS

16.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum gate, unitary transformation.

Physical gate fidelity remains an empirical engineering question.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Physical gate fidelity remains an empirical engineering question.

16.8 Exact evidence identities

- Source manifest: `sha256:42bb13c0646eea4cd1dbb2822a4cc213980cbb6bf622d108985528f6b540117f`
- Complete-census certificate: `sha256:9c9d3340df70fc9aa032104c6f83309f96ba6ce28721c8c3187e1dd29749e38e`
- Derivation seal: `sha256:d93e94a7c4c89f79904d0068dc37fb48db7040bb45533a02cb96f8c45cb617c1`
- Independent implementation: `sha256:629fa6869c3efd87e733736338d25b83adf646fe360ce5210876449f86aa18d5`
- Independent certificate: `sha256:8c737f9cb9e579c043660fde438e8405e4f6288e9d65f74e470be69d59549c89`
- External validation: `sha256:9c8ee6c1d374120eab1601d55d5d859b2391a968447e023a1f818301f7f114b7`
- Model-admitted engine receipt: `sha256:495704206fd6c701e1baf8d1ec0b4bcfd430a002aea0bb76453a091e01ccfdaf`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-GATE-001-495704206fd6c701.json`

17. Derivation 9: Fold quantum circuit syntax and semantics law

Claim: `SFT-QUANTUM-CIRCUIT-001`

17.1 Scientific necessity and exact theorem

Fold quantum circuit syntax and semantics law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced quantum circuits kernel retains generated wires; information units; reversible gates; causal circuit order; complete branchwise evaluation; inverse circuit; observation boundary; resource ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-GATE-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

17.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: generated wires; information units; reversible gates; causal circuit order; complete branchwise evaluation; inverse circuit; observation boundary; resource ledger.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum circuits.

> *All finite quantum circuits structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. This derives circuit semantics; hardware timing and noise are external.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

17.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.

Eliminated candidates	Exact first-failure reason
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced quantum circuits kernel retains generated wires; information units; reversible gates; causal circuit order; complete branchwise evaluation; inverse circuit; observation boundary; resource ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

17.4 Operational laws and consequences

- Support law: generated wires; information units; reversible gates; causal circuit order; complete branchwise evaluation; inverse circuit; observation boundary; resource ledger.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

17.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum circuits record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

17.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-CIRCUIT-001; result: PASS.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS

Control	Expected	Observed	Result
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; This derives circuit semantics; hardware timing and noise are external.	PASS

17.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum circuit, circuit semantics.

This derives circuit semantics; hardware timing and noise are external.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- This derives circuit semantics; hardware timing and noise are external.

17.8 Exact evidence identities

- Source manifest: `sha256:d4825331a10928d077303fda7e766c916140bf5346464aedd390634a321b7586`
- Complete-census certificate: `sha256:23cba4f14d69066c4576001a3d04ea7f054118d48815312b06b97cb9d14db74c`
- Derivation seal: `sha256:8fb9825eeec2daff4511f6aaf81f866fe575a23d070bbfbbef9d4d5d59837990`
- Independent implementation: `sha256:1d370c1f6d10c8b50b5f4ce8980a2557dead4382a9a3565990df366ab519fdd6`
- Independent certificate: `sha256:613d70fcb5954bd320bae00fe784bb03726d6ab1b3cc0c287a6df616d4746618`
- External validation: `sha256:0a859bd98ada50cc93ff7f774518d1fa57ec325f78cda38db2219c74eaa48ceb`
- Model-admitted engine receipt: `sha256:63e751cf9ab382cdc5c4b5dfd0c6c2f01ad90659bf2c9189357f141475d9e8aa`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-CIRCUIT-001-63e751cf9ab382cd.json`

18. Derivation 10: Fold quantum universality law

Claim: `SFT-QUANTUM-UNIVERSALITY-001`

18.1 Scientific necessity and exact theorem

Fold quantum universality law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced quantum universal computation kernel retains frozen gate-description grammar; exact encoded circuits; one interpreter; branchwise trace simulation; phase and observation preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-CIRCUIT-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

18.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: frozen gate-description grammar; exact encoded circuits; one interpreter; branchwise trace simulation; phase and observation preservation; overhead ledger.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum universal computation.

> *All finite quantum universal computation structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Universality is exact for every generated description in the frozen grammar.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

18.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced quantum universal computation kernel retains frozen gate-description grammar; exact encoded circuits; one interpreter; branchwise trace simulation; phase and observation preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

18.4 Operational laws and consequences

- Support law: frozen gate-description grammar; exact encoded circuits; one interpreter; branchwise trace simulation; phase and observation preservation; overhead ledger.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

18.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum universal computation record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

18.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Universality is exact for every generated description in the frozen grammar.	PASS

18.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: universal quantum computer, universal gate set.

Universality is exact for every generated description in the frozen grammar.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Universality is exact for every generated description in the frozen grammar.

18.8 Exact evidence identities

- Source manifest: `sha256:59bfa7dc2618cb23f1897460501e00e4fdd1f0a0b1fab31fc0ffc9e06de8c549`

- Complete-census certificate: `sha256:4af4128a42fc2e6efbb404ece4464adde28e77475392686199b67aa7a00117be`
- Derivation seal: `sha256:7f480fe1655f32f7ff3f640e9e0f508c501d21b0b44430f24ff41fcc84d35c0a`
- Independent implementation: `sha256:00dc7bb66a94f5a7f0ced9c5b08dea0b489c312288e3cc0bf250f86bc8a7d89f`
- Independent certificate: `sha256:0d501674df0431ac2f24be59fafd160e2c225807ab1b000013632f09ddcd1214`
- External validation: `sha256:43e4644556a5eb410b174b4cdc7d6993d965e7f1b5636773dd41e3c67ba148d1`
- Model-admitted engine receipt: `sha256:28b6a73e02450c70bf3ee49f3fdcdf2e28b30c463e05218b99cdc8cda723e86a`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-UNIVERSALITY-001-28b6a73e02450c70.json`

19. Derivation 11: Fold quantum algorithm-family law

Claim: `SFT-QUANTUM-ALGORITHMS-001`

19.1 Scientific necessity and exact theorem

Fold quantum algorithm-family law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced quantum algorithms kernel retains complete input support; reversible branch generation; phase action; interference merge; observation decoder; correctness trace per branch; classical comparison boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-UNIVERSALITY-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

19.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: complete input support; reversible branch generation; phase action; interference merge; observation decoder; correctness trace per branch; classical comparison boundary.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum algorithms.

> *All finite quantum algorithms structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Named speedups require their own exact problem family and resource proof.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

19.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced quantum algorithms kernel retains complete input support; reversible branch generation; phase action; interference merge; observation decoder; correctness trace per branch; classical comparison boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

19.4 Operational laws and consequences

- Support law: complete input support; reversible branch generation; phase action; interference merge; observation decoder; correctness trace per branch; classical comparison boundary.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

19.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum algorithms record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

19.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS

Control	Expected	Observed	Result
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Named speedups require their own exact problem family and resource proof.	PASS

19.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum algorithm, quantum speedup.

Named speedups require their own exact problem family and resource proof.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Named speedups require their own exact problem family and resource proof.

19.8 Exact evidence identities

- Source manifest: `sha256:779315c063473730553e1b94b47294d97acd1023e130e685d2aee25fd1c2ef67`
- Complete-census certificate: `sha256:63c84d1546a1f83c51991b85670ab300de70ad6532104733532a81dc371c7282`
- Derivation seal: `sha256:cb94fae7c67d50383d283b028c580568c6d7e1fafb4a919217df581529703a87`
- Independent implementation: `sha256:89bb200d9b5c23e863521d13d47d326f8df6f7e4ace5605cd4e62929121a8065`
- Independent certificate: `sha256:8eac37074eb7bcf0579cb97aedc91fa70d056d0828515070ae6df1f157b8d22c`
- External validation: `sha256:fe1b4e628eb13b88f9e401532d3f00a562cb8264986a662d0716f9bf766d6409`
- Model-admitted engine receipt: `sha256:a33b13375d76ebf039fcc5711c0810ab99ac0b9456295bc7881d76c94558a86b`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-ALGORITHMS-001-a33b13375d76ebf0.json`

20. Derivation 12: Fold quantum complexity law

Claim: `SFT-QUANTUM-COMPLEXITY-001`

20.1 Scientific necessity and exact theorem

Fold quantum complexity law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced quantum computational resources kernel retains canonical input size; gate count; causal depth; live branch support; query and communication rows; measurement records; lower and upper witness boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-ALGORITHMS-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

20.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: canonical input size; gate count; causal depth; live branch support; query and communication rows; measurement records; lower and upper witness boundary.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum computational resources.

> *All finite quantum computational resources structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. No arbitrary separation between conventional complexity classes is claimed without a separate certificate.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

20.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced quantum computational resources kernel retains canonical input size; gate count; causal depth; live branch support; query and communication rows; measurement records; lower and upper witness boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

20.4 Operational laws and consequences

- Support law: canonical input size; gate count; causal depth; live branch support; query and communication rows; measurement records; lower and upper witness boundary.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

20.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum computational resources record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

20.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No arbitrary separation between conventional complexity classes is claimed without a separate certificate.	PASS

20.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum complexity, BQP.

No arbitrary separation between conventional complexity classes is claimed without a separate certificate.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- No arbitrary separation between conventional complexity classes is claimed without a separate certificate.

20.8 Exact evidence identities

- Source manifest: `sha256:8eac4910a233e30d537d3cc6dd92cfa0af9524abcff1bb6dae3c57759bf0f57f`

- Complete-census certificate: `sha256:43fdd77acc2f1d28c1d909f806c2bf5d1bfd03d84effc0c51387915d78f83c24`
- Derivation seal: `sha256:2dabda188246d3553502bbfb24cebd18a4513e169b7291aa5ecbf9ca5fd66fe9`
- Independent implementation: `sha256:81b3d49b18170b958f74bfe4f4a5cbd47bb4b0712c750027a7554d3576523419`
- Independent certificate: `sha256:94b58ef5a03dc1e832cdf2ee0a5158ca6d8d9626c6d514f939aee795a7d0eee6`
- External validation: `sha256:15ad7c676afff75fe9d9d991b0843f388b9061ed86b67929008d295eef2a2cd1`
- Model-admitted engine receipt: `sha256:d6307d961273c2a383fecd3b919c56319b910bccb58dbbcf5402a2a76a9ca8b6`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-COMPLEXITY-001-d6307d961273c2a3.json`

21. Derivation 13: Fold quantum communication law

Claim: `SFT-QUANTUM-COMMUNICATION-001`

21.1 Scientific necessity and exact theorem

Fold quantum communication law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced quantum communication kernel retains sender and receiver supports; joint state composition; local transformations; channel relation; retained correlations; observation records; resource and loss ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-COMPLEXITY-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

21.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: sender and receiver supports; joint state composition; local transformations; channel relation; retained correlations; observation records; resource and loss ledger.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum communication.

> *All finite quantum communication structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Physical transmission capacity and security require registered media experiments.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

21.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced quantum communication kernel retains sender and receiver supports; joint state composition; local transformations; channel relation; retained correlations; observation records; resource and loss ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

21.4 Operational laws and consequences

- Support law: sender and receiver supports; joint state composition; local transformations; channel relation; retained correlations; observation records; resource and loss ledger.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

21.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum communication record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

21.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS

Control	Expected	Observed	Result
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Physical transmission capacity and security require registered media experiments.	`PASS`

21.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum communication, teleportation.

Physical transmission capacity and security require registered media experiments.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Physical transmission capacity and security require registered media experiments.

21.8 Exact evidence identities

- Source manifest: `sha256:8cbc3dcc0ee105c719a4c9019ba6f993d6bd51d32427286b18f5317b9a8ab96c`
- Complete-census certificate: `sha256:288320813c5011753ac302d9c4d463a58e58623ae71d344625eea30eafcafe70`
- Derivation seal: `sha256:2b321b0b8c520885563b2c770bed8e0ec83e5aa0720a05a2f679125c5d1f63cc`
- Independent implementation: `sha256:c98029cec76386f0bea9768e3536a822803d2c7fb1eb3e16616272c1e0e97bec`
- Independent certificate: `sha256:b8460f43cb576317854272866636f450a0a837fd34929ae4b947d681dd95eb91`
- External validation: `sha256:a91fd78dd1739e147a926a006477eb6fa4bef506490fedcedb86827c0b56fd91`
- Model-admitted engine receipt: `sha256:01059b1cbe86b894ae6137de5625e1dfcd1cd640ac5634c0ca2260977993b97a`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-COMMUNICATION-001-01059b1cbe86b894.json`

22. Derivation 14: Fold quantum coding law

Claim: `SFT-QUANTUM-CODING-001`

22.1 Scientific necessity and exact theorem

Fold quantum coding law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced quantum coding kernel retains logical branch support; redundant joint words; reversible encoder; declared error actions; disjoint syndrome supports; decoder and reconstruction trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-COMMUNICATION-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

22.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: logical branch support; redundant joint words; reversible encoder; declared error actions; disjoint syndrome supports; decoder and reconstruction trace.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum coding.

> *All finite quantum coding structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. The code theorem applies only to the registered generated error family.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

22.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced quantum coding kernel retains logical branch support; redundant joint words; reversible encoder; declared error actions; disjoint syndrome supports; decoder and reconstruction trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

22.4 Operational laws and consequences

- Support law: logical branch support; redundant joint words; reversible encoder; declared error actions; disjoint syndrome supports; decoder and reconstruction trace.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

22.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum coding record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

22.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-CODING-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-CODING-001; result: PASS.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-CODING-001; result: PASS.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-CODING-001; result: PASS.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-CODING-001; result: PASS.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-CODING-001; result: PASS.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-CODING-001; result: PASS.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-CODING-001; result: PASS.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; The code theorem applies only to the registered generated error family.	PASS

22.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum code, encoding.

The code theorem applies only to the registered generated error family.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- The code theorem applies only to the registered generated error family.

22.8 Exact evidence identities

- Source manifest: `sha256:85e7aabcdad3f77f050dd76d25a562d3156936ff78a453bbf1f62cfdf528e8a8`
- Complete-census certificate: `sha256:93f1b6cf418d33752789ee30da4e3691204d71062d6e49138d4a6f9a292996aa`
- Derivation seal: `sha256:ea64d9cb56689b01f32e45805b41e053214e1e665f68f83c9fac52f894205efb`

- Independent implementation: `sha256:3e3936d7ac16b4f76240d81294e3dab60dc4110611c44429cea46df1444590f2`
- Independent certificate: `sha256:736fdb0c5a526c7b542e4eaf8fdf98681aae185d448844b242b1f371688b7746`
- External validation: `sha256:e7785a525808797f14dba72e1e20acbd6dbb176c7799c1e71afca1bca34b271f`
- Model-admitted engine receipt: `sha256:b79afc51951dfbebb30cd70fba329965b700fa14a16fb45b5754568ec7407a3e`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-CODING-001-b79afc51951dfbeb.json`

23. Derivation 15: Fold quantum error-correction law

Claim: `SFT-QUANTUM-ERROR-CORRECTION-001`

23.1 Scientific necessity and exact theorem

Fold quantum error-correction law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced quantum error correction kernel retains forced repetition width 2t+1; exact fault-depth trace; all error masks through t; majority held-label decoder; syndrome ledger; recovery certificate; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-CODING-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

23.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: forced repetition width 2t+1; exact fault-depth trace; all error masks through t; majority held-label decoder; syndrome ledger; recovery certificate.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum error correction.

> *All finite quantum error correction structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. The completed census covers generated fault depths one, two and three and the successor law for every positive finite t.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

23.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced quantum error correction kernel retains forced repetition width 2t+1; exact fault-depth trace; all error masks through t; majority held-label decoder; syndrome ledger; recovery certificate; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

23.4 Operational laws and consequences

- Support law: forced repetition width 2t+1; exact fault-depth trace; all error masks through t; majority held-label decoder; syndrome ledger; recovery certificate.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

23.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum error correction record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

23.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-ERROR-CORRECTION-001; result: `PASS`.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-ERROR-CORRECTION-001; result: `PASS`.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-ERROR-CORRECTION-001; result: `PASS`.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-ERROR-CORRECTION-001; result: `PASS`.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-ERROR-CORRECTION-001; result: `PASS`.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-ERROR-CORRECTION-001; result: `PASS`.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-ERROR-CORRECTION-001; result: `PASS`.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-ERROR-CORRECTION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS

Control	Expected	Observed	Result
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; The completed census covers generated fault depths one, two and three and the successor law for every positive finite t.	PASS

23.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum error correction, repetition code.

The completed census covers generated fault depths one, two and three and the successor law for every positive finite t.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- The completed census covers generated fault depths one, two and three and the successor law for every positive finite t.

23.8 Exact evidence identities

- Source manifest: `sha256:5ac864ccd27449cb7c1a756b5ed53c4094709160f00cf0e8f19e58f01a20a2d9`
- Complete-census certificate: `sha256:9712241740820a883217d33d982426a39c21903950f917eafa9fefd38d970389`
- Derivation seal: `sha256:61cc113c2d425a9f9c9d2e13bb1a983bd94e88220a6d91a4f1815d41b5059ee6`
- Independent implementation: `sha256:f3e41dd58f96c808ce4cadab817f8a4ee86d6369ef8e6e43ddfaacc792071af9`
- Independent certificate: `sha256:721e7c933eb6f760790d3ee15cd54171a46f8ffa24c67f5b59e62872b12fb16b`
- External validation: `sha256:c5784b25becf3d48609c4a3447a3cff73cd2822418dd3800282d1948dbe6d3ef`
- Model-admitted engine receipt: `sha256:a4791dcbf8753c32e8304c5d5e80f9bf7f21a7356b6adb9422ea60d5eaf59656`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-ERROR-CORRECTION-001-a4791dcbf8753c32.json`

24. Derivation 16: Fold quantum fault-tolerance law

Claim: `SFT-QUANTUM-FAULT-TOLERANCE-001`

24.1 Scientific necessity and exact theorem

Fold quantum fault-tolerance law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced fault-tolerant quantum computation kernel retains encoded logical support; declared component faults; transversal or contained transformations; recovery after each location; malignant-fault boundary; induction over circuit locations; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-ERROR-CORRECTION-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

24.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: encoded logical support; declared component faults; transversal or contained transformations; recovery after each location; malignant-fault boundary; induction over circuit locations.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for fault-tolerant quantum computation.

> *All finite fault-tolerant quantum computation structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. No hardware threshold constant is claimed; unlimited thresholds remain dependent on a separately derived physical fault model.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

24.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced fault-tolerant quantum computation kernel retains encoded logical support; declared component faults; transversal or contained transformations; recovery after each location; malignant-fault boundary; induction over circuit locations; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

24.4 Operational laws and consequences

- Support law: encoded logical support; declared component faults; transversal or contained transformations; recovery after each location; malignant-fault boundary; induction over circuit locations.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

24.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact fault-tolerant quantum computation record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

24.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: `PASS`.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: `PASS`.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: `PASS`.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: `PASS`.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: `PASS`.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: `PASS`.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: `PASS`.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No hardware threshold constant is claimed; unlimited thresholds remain dependent on a separately derived physical fault model.	PASS

24.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: fault tolerance, threshold theorem.

No hardware threshold constant is claimed; unlimited thresholds remain dependent on a separately derived physical fault model.

Version 1.1 reconciliation: the sealed 001 wording uses 'unlimited thresholds' for physical hardware thresholds. It must not be read as leaving the exact code-order theorem open. The separately generated and admitted `SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002` closes every supplied positive finite fault order by the forced width `2t+1`, while stochastic device thresholds remain empirical.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- No hardware threshold constant is claimed; unlimited thresholds remain dependent on a separately derived physical fault model.

24.8 Exact evidence identities

- Source manifest: `sha256:3260e928de52083476ff03669bb6321999c2e7042d32a55027e98040c2d16c4e`
- Complete-census certificate: `sha256:1e9b2a7fc565525f5353d4590607e2f6d1450e18f5c3db873f9f8b61c229960f`
- Derivation seal: `sha256:1816e6b491362e7cc5b5d306e833227bdf765559d55693d0b18553e2911f1a57`
- Independent implementation: `sha256:ef9893933f17387358f0d7512e05451d206c5ce7a11fe348bbc1052bd42ee0dd`
- Independent certificate: `sha256:0a28a0751ef1e16a701eb0fdf84d8d4a2583c0e4314f9cd1d4badfd2e5a10c7a`
- External validation: `sha256:367a4648636683bd04852d04c27eb6360aede4d53948a7002d5d0feea0a37ea2`
- Model-admitted engine receipt: `sha256:9f2526c5bd9f1625a3cee65b0cba4e9a600ecdc9a8031ac51367e35c37308b16`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-FAULT-TOLERANCE-001-9f2526c5bd9f1625.json`

25. Derivation 17: Fold quantum simulation law

Claim: `SFT-QUANTUM-SIMULATION-001`

25.1 Scientific necessity and exact theorem

Fold quantum simulation law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced quantum simulation kernel retains encoded target state support; reversible update law; phase and interference traces; observation map; exact simulator correspondence; resource ledger; falsification boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-FAULT-TOLERANCE-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`

- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

25.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: encoded target state support; reversible update law; phase and interference traces; observation map; exact simulator correspondence; resource ledger; falsification boundary.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum simulation.

> *All finite quantum simulation structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. A simulator derives consequences of its registered model and cannot establish a natural quantum law by itself.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

25.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.

Eliminated candidates	Exact first-failure reason
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced quantum simulation kernel retains encoded target state support; reversible update law; phase and interference traces; observation map; exact simulator correspondence; resource ledger; falsification boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

25.4 Operational laws and consequences

- Support law: encoded target state support; reversible update law; phase and interference traces; observation map; exact simulator correspondence; resource ledger; falsification boundary.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

25.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum simulation record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

25.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-SIMULATION-001; result: PASS.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; A simulator derives consequences of its registered model and cannot establish a natural quantum law by itself.	PASS

25.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum simulation, Hamiltonian simulation.

A simulator derives consequences of its registered model and cannot establish a natural quantum law by itself.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- A simulator derives consequences of its registered model and cannot establish a natural quantum law by itself.

25.8 Exact evidence identities

- Source manifest: `sha256:a376c3321387216b559a5d2543ece347699b17421f174831037f93ee4012ba23`
- Complete-census certificate: `sha256:0036a9974eaf22533d40dbeffc708a9b259140447ebb99dfd5406040e63c341b`
- Derivation seal: `sha256:bb1d2c15251c769fbcf1692edc85f6253cb6f103dbffe04680cedb4d492cbff8`
- Independent implementation: `sha256:72f00440c8b0cfeaa26940b8aef2728b3e74896b435207b9b8f9426348064df9`
- Independent certificate: `sha256:fcdb1c83f2a12bee60d57319cf49492523c03723aee2e0aafa22e3a88abff900`
- External validation: `sha256:2e19399541f4035708283c1c043842f1c5b8ffef1330ec004109a14ff3fdfbd0`
- Model-admitted engine receipt: `sha256:daf61c89738fb9f6370bc244c3a7538a3c4037ab408dfd5d15e3301c57e817a6`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-SIMULATION-001-daf61c89738fb9f6.json`

26. Derivation 18: Fold quantum verification law

Claim: `SFT-QUANTUM-VERIFICATION-001`

26.1 Scientific necessity and exact theorem

Fold quantum verification law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced quantum verification kernel retains claim and witness supports; verifier circuit; challenge schedule; branchwise acceptance; soundness and completeness ledgers; measurement records; tamper controls; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-SIMULATION-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

26.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: claim and witness supports; verifier circuit; challenge schedule; branchwise acceptance; soundness and completeness ledgers; measurement records; tamper controls.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum verification.

> *All finite quantum verification structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Security and correctness are exact only for the registered prover and resource grammar.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

26.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced quantum verification kernel retains claim and witness supports; verifier circuit; challenge schedule; branchwise acceptance; soundness and completeness ledgers; measurement records; tamper controls; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

26.4 Operational laws and consequences

- Support law: claim and witness supports; verifier circuit; challenge schedule; branchwise acceptance; soundness and completeness ledgers; measurement records; tamper controls.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

26.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum verification record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

26.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-VERIFICATION-001; result: PASS.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Security and correctness are exact only for the registered prover and resource grammar.	PASS

26.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum verification, QMA.

Security and correctness are exact only for the registered prover and resource grammar.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum

- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Security and correctness are exact only for the registered prover and resource grammar.

26.8 Exact evidence identities

- Source manifest: `sha256:d17c41d57b8e1a861e03983b2a5a7808ec8756e28c7361cb95bb1f5a159635ee`
- Complete-census certificate: `sha256:c54587141b8c1a489ee2b3d83d53548c435f0746abda0ab11358bbacb1fbe8ef`
- Derivation seal: `sha256:62b1704737e141ef52eab6b9687ec06ab6de072c1ba2dd8cc678880fe50e3ce0`
- Independent implementation: `sha256:00bb831c0de419ea512cd89dd478e15c3804c348f3b094de0d7321006f9f408f`
- Independent certificate: `sha256:fa884fd0ca86e8ae875b76054696f32cd5ef731fbbd1986b9a02b84140ece03a`
- External validation: `sha256:753bc09d36d4c45faa75aa9f02e0780bc6c49a45126630495deb18d72c74cd77`
- Model-admitted engine receipt: `sha256:6ca70e1a9487b428bc6e04e5e3b1a11907b79d1ed92f7d03a013a49380ecce54`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-VERIFICATION-001-6ca70e1a9487b428.json`

27. Derivation 19: Fold quantum learning law

Claim: `SFT-QUANTUM-LEARNING-001`

27.1 Scientific necessity and exact theorem

Fold quantum learning law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced quantum learning kernel retains classical or joint example support; reversible hypothesis process; phase/interference search; observation decoder; retained alternatives; sample and gate resources; sealed evaluation; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-VERIFICATION-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

27.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: classical or joint example support; reversible hypothesis process; phase/interference search; observation decoder; retained alternatives; sample and gate resources; sealed evaluation.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum learning.

> *All finite quantum learning structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. No pretrained quantum model, fitted amplitude or unexplained speedup is admitted.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

27.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced quantum learning kernel retains classical or joint example support; reversible hypothesis process; phase/interference search; observation decoder; retained alternatives; sample and gate resources; sealed evaluation; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

27.4 Operational laws and consequences

- Support law: classical or joint example support; reversible hypothesis process; phase/interference search; observation decoder; retained alternatives; sample and gate resources; sealed evaluation.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

27.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum learning record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

27.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-LEARNING-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-LEARNING-001; result: PASS.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-LEARNING-001; result: PASS.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-LEARNING-001; result: PASS.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-LEARNING-001; result: PASS.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-LEARNING-001; result: PASS.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-LEARNING-001; result: PASS.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-LEARNING-001; result: PASS.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS

Control	Expected	Observed	Result
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No pretrained quantum model, fitted amplitude or unexplained speedup is admitted.	PASS

27.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum learning, quantum machine learning.

No pretrained quantum model, fitted amplitude or unexplained speedup is admitted.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- No pretrained quantum model, fitted amplitude or unexplained speedup is admitted.

27.8 Exact evidence identities

- Source manifest: `sha256:720947592f997269012cdac1186615d307e7b3a553f0ff0d1e42eac0daedd840`
- Complete-census certificate: `sha256:4f049e2fe5f2c3499b0b6012f93625016627146fccbfc47dc3bc5dfcca22831c`
- Derivation seal: `sha256:1d75a66403b8bd208fad0ca8cdd0de260f54865c8eb4e1b1984a25888845b435`
- Independent implementation: `sha256:de8da3d2eb9a1c0c7cf06c6f851c311161c5e967d182c53c1d2d4f61d54045e3`
- Independent certificate: `sha256:301937de6bfe10c6fc95c62a92efa37c27d5866795d316932652ba6dd953bd5d`
- External validation: `sha256:2b0d54edaf033b569004c5d6e52026adaf3d9d5e8ad1129ff7ec246db2cebb42`
- Model-admitted engine receipt: `sha256:c4388e69ff7412fe09e53b7e4b2a4eda12168ff6665ca351758c185ac7e018b7`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-LEARNING-001-c4388e69ff7412fe.json`

28. Derivation 20: Full operational classical-quantum correspondence law

Claim: `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`

28.1 Scientific necessity and exact theorem

Full operational classical-quantum correspondence law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced classical and quantum operational correspondence kernel retains common generated descriptions; embedding of classical states; reversible classical submodel; branchwise quantum execution; measurement decoder; bidirectional result preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-LEARNING-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

28.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: common generated descriptions; embedding of classical states; reversible classical submodel; branchwise quantum execution; measurement decoder; bidirectional result preservation; overhead ledger.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for classical and quantum operational correspondence.

> *All finite classical and quantum operational correspondence structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Correspondence does not identify the models where phase-sensitive joint support changes the operational trace.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

28.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced classical and quantum operational correspondence kernel retains common generated descriptions; embedding of classical states; reversible classical submodel; branchwise quantum execution; measurement decoder; bidirectional result preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

28.4 Operational laws and consequences

- Support law: common generated descriptions; embedding of classical states; reversible classical submodel; branchwise quantum execution; measurement decoder; bidirectional result preservation; overhead ledger.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

28.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact classical and quantum operational correspondence record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

28.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Correspondence does not identify the models where phase-sensitive joint support changes the operational trace.	PASS

28.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: classical-quantum correspondence, deferred measurement.

Correspondence does not identify the models where phase-sensitive joint support changes the operational trace.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity

- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Correspondence does not identify the models where phase-sensitive joint support changes the operational trace.

28.8 Exact evidence identities

- Source manifest: `sha256:a784ae4b69af81c489a660162333bca6e55fb56d401856af1d8e14516e863595`
- Complete-census certificate: `sha256:922645a8e17e57036ddadbb6d3a35bf1e0a4a691864cb54ddfac3fb3f0978b59`
- Derivation seal: `sha256:d10d9d064e2a63254c2cae2d5d7ea26173427927650eb375fbf698f69cf61f86`
- Independent implementation: `sha256:1cd0dd23ea4b5b984d9a1ebe768005d0c09258970fd241aa5a02215cfdae7d2b`
- Independent certificate: `sha256:74d501babc031290bfc0b5a041bed885d7ccb65d700b20676a8260f75fd91519`
- External validation: `sha256:4c2be0efb0edeb4392351b788cd58b9331c850577e49ae1b0699ad2ad420445f`
- Model-admitted engine receipt: `sha256:e1712589e1aba3ce6536a3ba33d599e06bdf793a335079d1a775975bc234b614`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001-e1712589e1aba3ce.json`

29. Derivation 21: Limits of Fold quantum computation law

Claim: `SFT-QUANTUM-LIMITS-001`

29.1 Scientific necessity and exact theorem

Limits of Fold quantum computation law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

> *The forced limits of quantum computation kernel retains classical computability dependency; finite description grammar; exact quantum execution; self-description support; halting and undecidability transfer; resource and observation boundaries; no hypercomputational oracle; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001`
- `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`
- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-INFO-QUANTUM-CORRESPONDENCE-001`

29.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: classical computability dependency; finite description grammar; exact quantum execution; self-description support; halting and undecidability transfer; resource and observation boundaries; no hypercomputational oracle.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for limits of quantum computation.

> *All finite limits of quantum computation structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Quantum computation does not escape the admitted computability and self-reference boundaries; physical claims remain empirical.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`partial-or-selected-branch-support`	partial-or-selected-branch-support removes the required exact support structure.	`complete-generated-branch-support`	complete-generated-branch-support retains the required exact support structure.
`unit`	`imported-bit-or-qubit`	imported-bit-or-qubit removes the required exact unit structure.	`one-Fold-distinction-unit`	one-Fold-distinction-unit retains the required exact unit structure.
`composition`	`sampled-joint-states`	sampled-joint-states removes the required exact composition structure.	`complete-pair-cell-joint-support`	complete-pair-cell-joint-support retains the required exact composition structure.
`phase`	`complex-amplitude-postulate`	complex-amplitude-postulate removes the required exact phase structure.	`period-held-phase-action`	period-held-phase-action retains the required exact phase structure.
`transformation`	`partial-or-irreversible-map`	partial-or-irreversible-map removes the required exact transformation structure.	`total-reversible-source-bound-map`	total-reversible-source-bound-map retains the required exact transformation structure.
`observation`	`collapse-without-record`	collapse-without-record removes the required exact observation structure.	`retained-class-and-complete-record`	retained-class-and-complete-record retains the required exact observation structure.
`generality`	`fixed-circuit-examples`	fixed-circuit-examples removes the required exact generality structure.	`branch-and-depth-successor-certificate`	branch-and-depth-successor-certificate retains the required exact generality structure.
`addition`	`imported-quantum-model`	imported-quantum-model removes the required exact addition structure.	`no-extra-quantum-premise`	no-extra-quantum-premise retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

29.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

The unique all-preserving member is `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`.

> *The forced limits of quantum computation kernel retains classical computability dependency; finite description grammar; exact quantum execution; self-description support; halting and undecidability transfer; resource and observation boundaries; no hypercomputational oracle; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

29.4 Operational laws and consequences

- Support law: classical computability dependency; finite description grammar; exact quantum execution; self-description support; halting and undecidability transfer; resource and observation boundaries; no hypercomputational oracle.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

29.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact limits of quantum computation record.

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

29.6 Executed witnesses and adverse controls

- `quantum-witness-1` - exact Fold quantum operational witness 1 for SFT-QUANTUM-LIMITS-001; result: PASS.
- `quantum-witness-2` - exact Fold quantum operational witness 2 for SFT-QUANTUM-LIMITS-001; result: PASS.
- `quantum-witness-3` - exact Fold quantum operational witness 3 for SFT-QUANTUM-LIMITS-001; result: PASS.
- `quantum-witness-4` - exact Fold quantum operational witness 4 for SFT-QUANTUM-LIMITS-001; result: PASS.
- `quantum-witness-5` - exact Fold quantum operational witness 5 for SFT-QUANTUM-LIMITS-001; result: PASS.
- `quantum-witness-6` - exact Fold quantum operational witness 6 for SFT-QUANTUM-LIMITS-001; result: PASS.
- `quantum-witness-7` - exact Fold quantum operational witness 7 for SFT-QUANTUM-LIMITS-001; result: PASS.
- `quantum-witness-8` - exact Fold quantum operational witness 8 for SFT-QUANTUM-LIMITS-001; result: PASS.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS

Control	Expected	Observed	Result
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Quantum computation does not escape the admitted computability and self-reference boundaries; physical claims remain empirical.	PASS

29.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: limits of quantum computing, quantum computability.

Quantum computation does not escape the admitted computability and self-reference boundaries; physical claims remain empirical.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Quantum computation does not escape the admitted computability and self-reference boundaries; physical claims remain empirical.

29.8 Exact evidence identities

- Source manifest: `sha256:f111f69d04fe049625cc50c9378bfca7d08fd3c2e6cc82cfbc7151cf183223db`
- Complete-census certificate: `sha256:d9059ae5cf371742769379c802668b1bc83493f42cf636184e5357c3e3279f95`
- Derivation seal: `sha256:4abd56778c0176f6426706806e75fdff866dedbbe2e78345585c0f9d0c6f360f`
- Independent implementation: `sha256:f1ed114f18db0541db98fc2966784fabca2116c53921fb473b68c2e866a204c5`
- Independent certificate: `sha256:e018eb19348a191249d0fa680329d0cf71d1c2e1d62b0734732426f74afb2cf5`
- External validation: `sha256:2eaff27d824980b4d9b2b6cfade5b6a24002915fd4faa1ba47c521331707c176`
- Model-admitted engine receipt: `sha256:332e3b1f6df5725ae3e5cf97d9080daa26fd501e7138f1b9d1c7b4837469c545`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-LIMITS-001-332e3b1f6df5725a.json`

30. Derivation 22: Unbounded finite Fold quantum fault-tolerance law

Claim: `SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002`

30.1 Scientific necessity and exact theorem

The later V2 result removes the fixed-order frontier and requires a constructive depth-independent theorem for every supplied positive finite fault allowance.

The engine-admitted theorem is:

> *For every supplied positive finite fault-order trace t, the unique first repetition width correcting every mask through t is 2t+1: at least t+1 source labels remain against at most t changes, every width through 2t has a constructive tie or wrong-majority counterexample, redundancy is exactly 2t, and the successor fault order adds exactly two labels. The corrected word composes with exact Fold circuit semantics. This is an unbounded finite fault-order theorem, not a stochastic hardware-rate or physical threshold constant.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-QUANTUM-ERROR-CORRECTION-001`
- `SFT-QUANTUM-FAULT-TOLERANCE-001`
- `SFT-QUANTUM-CIRCUIT-001`
- `SFT-COMP-CPLX-ARBITRARY-CIRCUIT-LOWER-BOUND-002`

30.2 Generated grammar and exact boundary

Strict-majority preservation forces the width: t changed positions require strictly more than t retained source positions, so the least total is t+(t+1)=2t+1. The generated counterexample at every predecessor width proves minimality, and the fault-order successor proves unbounded positive-finite closure.

Generation rule: Generate the literal product of fault order, width, mask coverage, minimality, recovery, circuit composition, successor and no-extra-parameter coordinates.

> *Every supplied positive finite Fold fault-order trace, its generated repetition widths in increasing order, every fault mask through that order by exact subset class, and composition with admitted finite Fold circuit semantics.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`fault_order`	`fixed-one-error-model`	fixed-one-error-model removes the required exact fault_order structure.	`supplied-positive-finite-fault-trace`	supplied-positive-finite-fault-trace retains the required exact fault_order structure.
`width`	`selected-or-even-width`	selected-or-even-width removes the required exact width structure.	`first-strict-majority-width-2t-plus-1`	first-strict-majority-width-2t-plus-1 retains the required exact width structure.
`coverage`	`sampled-fault-patterns`	sampled-fault-patterns removes the required exact coverage structure.	`all-masks-through-t-by-subset-count`	all-masks-through-t-by-subset-count retains the required exact coverage structure.
`minimality`	`shorter-widths-unchecked`	shorter-widths-unchecked removes the required exact minimality structure.	`counterexample-for-every-width-through-2t`	counterexample-for-every-width-through-2t retains the required exact minimality structure.
`recovery`	`decoded-label-without-record`	decoded-label-without-record removes the required exact recovery structure.	`exact-syndrome-and-recovery-trace`	exact-syndrome-and-recovery-trace retains the required exact recovery structure.
`circuit`	`isolated-code-example`	isolated-code-example removes the required exact circuit structure.	`corrected-word-feeds-exact-circuit-semantics`	corrected-word-feeds-exact-circuit-semantics retains the required exact circuit structure.
`generality`	`fixed-t-table`	fixed-t-table removes the required exact generality structure.	`fault-order-successor-and-ceiling-defeat`	fault-order-successor-and-ceiling-defeat retains the required exact generality structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`addition`	`imported-hardware-threshold`	imported-hardware-threshold removes the required exact addition structure.	`no-rate-device-or-noise-parameter`	no-rate-device-or-noise-parameter retains the required exact addition structure.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

30.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	fixed-one-error-model removes the required exact fault_order structure.
64	selected-or-even-width removes the required exact width structure.
32	sampled-fault-patterns removes the required exact coverage structure.
16	shorter-widths-unchecked removes the required exact minimality structure.
8	decoded-label-without-record removes the required exact recovery structure.
4	isolated-code-example removes the required exact circuit structure.
2	fixed-t-table removes the required exact generality structure.
1	imported-hardware-threshold removes the required exact addition structure.

The unique all-preserving member is `supplied-positive-finite-fault-trace__first-strict-majority-width-2t-plus-1__all-masks-through-t-by-subset-count__counterexample-for-every-width-through-2t__exact-syndrome-and-recovery-trace__corrected-word-feeds-exact-circuit-semantics__fault-order-successor-and-ceiling-defeat__no-rate-device-or-noise-parameter`.

> *For every supplied positive finite fault-order trace t, the unique first repetition width correcting every mask through t is 2t+1: at least t+1 source labels remain against at most t changes, every width through 2t has a constructive tie or wrong-majority counterexample, redundancy is exactly 2t, and the successor fault order adds exactly two labels. The corrected word composes with exact Fold circuit semantics. This is an unbounded finite fault-order theorem, not a stochastic hardware-rate or physical threshold constant.*

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

30.4 Operational laws and consequences

- Width law: the first strict-majority carrier at fault order t contains exactly 2t+1 held positions.
- Correction law: at most t changes leave at least t+1 original labels, so the unique strict majority is retained.
- Minimality law: every width w through 2t is defeated by changing ceil(w/2) positions, producing a tie or wrong majority.
- Successor law: replacing t by its next generated fault order adds exactly two carrier positions and defeats every fixed positive finite ceiling.
- Composition law: recovery supplies the exact source word to the already-admitted reversible circuit semantics.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

30.5 Depth-independent closure

Base: At the first positive fault order, widths one and two have explicit failure rows and width three retains two source labels against one change.

Successor: If width 2t+1 is the first survivor at fault order t, the next order requires one further possible change and one further retained source majority; exactly two added positions produce 2(t+1)+1, while every shorter successor width has the same constructive counterexample.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

30.6 Executed witnesses and adverse controls

- `unbounded-fault-witness-1` - one-error order forces width three for SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002; result: PASS.
- `unbounded-fault-witness-2` - separate two- and three-error orders force widths five and seven for SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002; result: PASS.
- `unbounded-fault-witness-3` - fault order fourteen forces width twenty-nine for SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002; result: PASS.
- `unbounded-fault-witness-4` - every mask through orders one, two and three decodes exactly for SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002; result: PASS.
- `unbounded-fault-witness-5` - every predecessor width through fault order fourteen has a constructive failure for SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002; result: PASS.
- `unbounded-fault-witness-6` - the fault-order successor adds exactly two labels for SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002; result: PASS.
- `unbounded-fault-witness-7` - every supplied finite ceiling has a generated successor fault order for SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002; result: PASS.
- `unbounded-fault-witness-8` - all depth-seven words round-trip at fault order fourteen before circuit evaluation for SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002; result: PASS.

Control	Expected	Observed	Result
`false_premise`	reject incomplete state support	fixed-one-error-model removes the required exact fault_order structure.	PASS
`tampered_source`	reject changed source identity	changed source differs	PASS
`tampered_artifact`	reject altered survivor support	one literal survivor	PASS
`boundary`	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinite-width code or completed infinity; no measured error rate, hardware threshold, fitted noise model or device parameter; no application output or physical benchmark may select the code law	PASS

30.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum error correction, fault tolerance, threshold theorem boundary.

The theorem is unbounded over supplied positive finite code fault orders. It does not estimate a stochastic physical device threshold, correlated hardware noise rate or thermodynamic implementation cost.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinite-width code or completed infinity
- no measured error rate, hardware threshold, fitted noise model or device parameter
- no application output or physical benchmark may select the code law

30.8 Exact evidence identities

- Source manifest: `sha256:46d41cc55c7f4d0fba44e1a189bebcab0183453b75f53b24e8fd9a8959e6996e`
- Complete-census certificate: `sha256:64561c32af9d020a6344cf3c2b92a2a332ee102671e713ead0acfe8e10bdc58b`
- Derivation seal: `sha256:3f4c7dd00b70b80685d2182f1c66afb4d2ed6ff33d7fb2e1e0c574afeaffd2e3`
- Independent implementation: `sha256:08f284371749dfe9c00b2f28d468d9006d7cebfccd62ec6d6fa8a54b4d1b45d2`
- Independent certificate: `sha256:8260f560843df345b17ec1cc8d7ff891a3486f59dfcdf9d28fa9a1313e9fed93`
- External validation: `sha256:64d49f73d9b50bee7174034718386bbe89ed8d292ee33fe37580a61bdf55acf3`
- Model-admitted engine receipt: `sha256:97e061e60929c595adbc3706113e920f80a7b9ea224b3606893edd4c244d445b`
- Receipt path: `receipts/engine/model_admitted/SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002-97e061e60929c595.json`

31. Branch synthesis

The 22 laws establish one operational machine family in which classical and quantum modes share descriptions, traces, resources and verification. Quantum distinctions arise from complete branch support, held phase action, predecessor merging and nonfactorable composition; they are not imported as a separate mathematical universe.

32. Complete V1/V2 reconstruction audit

The categorical-owner audit reads all 763 V1/V2 census entries. It identifies 9 formal Quantum Computation source entries, decomposes them into 29 obligations, closes 29 at same strength and leaves 0 open. V1's quantum-named natural phenomena are not omitted: they are explicitly assigned to Physics because physical quantum effects and measured constants are not formal computation laws.

Source	Entry	Atomic obligation	Disposition	V3 receipt
V2	351	Same-strength reconstruction of Fold quantum complexity law: The forced quantum computational resources kernel retains canonical input size; gate count; causal depth; live branch support; query and communication rows; measurement records; lower and upper witness boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	`closed`	`SFT-QUANTUM-COMPLEXITY-001`

Source	Entry	Atomic obligation	Disposition	V3 receipt
V2	372	Same-strength reconstruction of Fold quantum algorithm-family law: The forced quantum algorithms kernel retains complete input support; reversible branch generation; phase action; interference merge; observation decoder; correctness trace per branch; classical comparison boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-ALGORITHMS-001
V2	397	Same-strength reconstruction of Complete Fold reversible-computation model: The forced reversible computation kernel retains canonical configurations; bijective transition rows; retained held labels; exact inverse traces; universal reversible interpretation; complete resource ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-REVERSIBLE-MODEL-001
V2	397	Same-strength reconstruction of Fold quantum information-unit law: The forced quantum information units kernel retains one Fold distinction; two held fibre labels; complete word support; exact observation classes; retained phase-label carrier; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-INFORMATION-UNIT-001

Source	Entry	Atomic obligation	Disposition	V3 receipt
V2	397	Same-strength reconstruction of Fold quantum state-composition law: The forced quantum state composition kernel retains complete branch support; exact pair-cell joint words; component identities; phase-label composition; marginal reconstruction; product and correlated classes; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	`closed`	`SFT-QUANTUM-STATE-COMPOSITION-001`
V2	397	Same-strength reconstruction of Superposition-equivalent Fold support law: The forced superposition-equivalent structure kernel retains complete b^k Fold-word support; one record per branch; held phase label per branch; no selected actual branch before observation; exact support provenance; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	`closed`	`SFT-QUANTUM-SUPERPOSITION-001`
V2	397	Same-strength reconstruction of Fold phase and interference operation law: The forced phase and interference kernel retains period-b held phase cycle; reversible phase action; total branch-to-image relation; complete predecessor fibres; same-image merge ledger; retained phase provenance; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	`closed`	`SFT-QUANTUM-PHASE-INTERFERENCE-001`

Source	Entry	Atomic obligation	Disposition	V3 receipt
V2	397	Same-strength reconstruction of Fold entangling-composition law: The forced entanglement kernel retains joint word support; exact marginals; product-support test; nonfactorable pair-cell classes; retained component observations; compositional trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-ENTANGLEMENT-001
V2	397	Same-strength reconstruction of Fold quantum measurement-semantics law: The forced measurement kernel retains total observation relation; retained observation class; closed branch distinctions; complete pre-observation record; selected-class state; exact reconstruction boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-MEASUREMENT-001
V2	398	Same-strength reconstruction of Fold reversible transformation and gate law: The forced quantum transformations and gates kernel retains bijective branch-word map; period-held phase action; complete domain and image support; exact inverse; source-bound gate identity; composition interface; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-GATE-001

Source	Entry	Atomic obligation	Disposition	V3 receipt
V2	398	Same-strength reconstruction of Fold quantum circuit syntax and semantics law: The forced quantum circuits kernel retains generated wires; information units; reversible gates; causal circuit order; complete branchwise evaluation; inverse circuit; observation boundary; resource ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	`closed`	`SFT-QUANTUM-CIRCUIT-001`
V2	398	Same-strength reconstruction of Fold quantum universality law: The forced quantum universal computation kernel retains frozen gate-description grammar; exact encoded circuits; one interpreter; branchwise trace simulation; phase and observation preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	`closed`	`SFT-QUANTUM-UNIVERSALITY-001`
V2	398	Same-strength reconstruction of Fold quantum algorithm-family law: The forced quantum algorithms kernel retains complete input support; reversible branch generation; phase action; interference merge; observation decoder; correctness trace per branch; classical comparison boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	`closed`	`SFT-QUANTUM-ALGORITHMS-001`

Source	Entry	Atomic obligation	Disposition	V3 receipt
V2	398	Same-strength reconstruction of Fold quantum complexity law: The forced quantum computational resources kernel retains canonical input size; gate count; causal depth; live branch support; query and communication rows; measurement records; lower and upper witness boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-COMPLEXITY-001
V2	399	Same-strength reconstruction of Fold quantum communication law: The forced quantum communication kernel retains sender and receiver supports; joint state composition; local transformations; channel relation; retained correlations; observation records; resource and loss ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-COMMUNICATION-001
V2	399	Same-strength reconstruction of Fold quantum coding law: The forced quantum coding kernel retains logical branch support; redundant joint words; reversible encoder; declared error actions; disjoint syndrome supports; decoder and reconstruction trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-CODING-001

Source	Entry	Atomic obligation	Disposition	V3 receipt
V2	399	Same-strength reconstruction of Fold quantum error-correction law: The forced quantum error correction kernel retains forced repetition width 2t+1; exact fault-depth trace; all error masks through t; majority held-label decoder; syndrome ledger; recovery certificate; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-ERROR-CORRECTION-001
V2	399	Same-strength reconstruction of Fold quantum fault-tolerance law: The forced fault-tolerant quantum computation kernel retains encoded logical support; declared component faults; transversal or contained transformations; recovery after each location; malignant-fault boundary; induction over circuit locations; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-FAULT-TOLERANCE-001
V2	400	Same-strength reconstruction of Fold quantum simulation law: The forced quantum simulation kernel retains encoded target state support; reversible update law; phase and interference traces; observation map; exact simulator correspondence; resource ledger; falsification boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-SIMULATION-001

Source	Entry	Atomic obligation	Disposition	V3 receipt
V2	400	Same-strength reconstruction of Fold quantum verification law: The forced quantum verification kernel retains claim and witness supports; verifier circuit; challenge schedule; branchwise acceptance; soundness and completeness ledgers; measurement records; tamper controls; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-VERIFICATION-001
V2	400	Same-strength reconstruction of Fold quantum learning law: The forced quantum learning kernel retains classical or joint example support; reversible hypothesis process; phase/interference search; observation decoder; retained alternatives; sample and gate resources; sealed evaluation; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-LEARNING-001
V2	400	Same-strength reconstruction of Full operational classical-quantum correspondence law: The forced classical and quantum operational correspondence kernel retains common generated descriptions; embedding of classical states; reversible classical submodel; branchwise quantum execution; measurement decoder; bidirectional result preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001

Source	Entry	Atomic obligation	Disposition	V3 receipt
V2	400	Same-strength reconstruction of Limits of Fold quantum computation law: The forced limits of quantum computation kernel retains classical computability dependency; finite description grammar; exact quantum execution; self-description support; halting and undecidability transfer; resource and observation boundaries; no hypercomputational oracle; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-LIMITS-001
V2	402	The complete b-way quantum support successor is reconstructed at every supplied finite depth.	closed	SFT-QUANTUM-SUPERPOSITION-001
V2	402	Quantum circuit support, size, width and depth retain exact base/successor resource recurrences.	closed	SFT-QUANTUM-CIRCUIT-001, SFT-QUANTUM-COMPLEXITY-001
V2	402	Each fresh quantum observation or reversible Fold layer adds its exact retained inverse record.	closed	SFT-QUANTUM-REVERSIBLE-MODEL-001, SFT-QUANTUM-MEASUREMENT-001
V2	403	Same-strength reconstruction of Fold quantum error-correction law: The forced quantum error correction kernel retains forced repetition width 2t+1; exact fault-depth trace; all error masks through t; majority held-label decoder; syndrome ledger; recovery certificate; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-ERROR-CORRECTION-001
V2	403	Same-strength reconstruction of Fold quantum fault-tolerance law: The forced fault-tolerant quantum computation kernel retains encoded logical support; declared component faults; transversal or contained transformations; recovery after each location; malignant-fault boundary; induction over circuit locations; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-FAULT-TOLERANCE-001

Source	Entry	Atomic obligation	Disposition	V3 receipt
V2	407	Same-strength reconstruction of Unbounded finite Fold quantum fault-tolerance law: For every supplied positive finite fault-order trace t, the unique first repetition width correcting every mask through t is 2t+1: at least t+1 source labels remain against at most t changes, every width through 2t has a constructive tie or wrong-majority counterexample, redundancy is exactly 2t, and the successor fault order adds exactly two labels. The corrected word composes with exact Fold circuit semantics. This is an unbounded finite fault-order theorem, not a stochastic hardware-rate or physical threshold constant.	closed	SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002

The audit also corrects the former wording around 'unlimited thresholds'. The current theorem is unbounded over supplied positive finite code fault orders. A measured stochastic hardware threshold is a different empirical object; it is neither denied nor claimed here and must be derived and tested in its correct Physics or engineering branch.

33. Quantum knowledge, open science and institutional accountability

Quantum science is especially vulnerable to authority by opacity: proprietary devices, inaccessible calibration data, result-only cloud interfaces and mathematical notation presented without the full route from premise to observation. Such systems may be useful instruments, but a reported score or vendor label cannot close a law when reviewers cannot reconstruct the support, transformation, measurement record, alternatives and failure boundary. This release exposes each of those objects.

The institutional critique is evidential, not an allegation that every funded scientist or institution acts in bad faith. Reviews report associations between commercial sponsorship and favorable outcomes, influence on research agendas, limits in grant-review reliability and fairness, and underpublication of null results. Capital, prestige, paywalls and consensus can therefore shape what becomes visible; none substitutes for an inspectable branch trace and independently reproducible receipt.

Maria Smith developed this programme without conventional academic credentials, appointment or a funding gate. That biography proves no quantum theorem; the public derivations must bear the whole scientific burden. It demonstrates why credentials cannot be permission to create knowledge and why every excluded mind is a scientific loss. Expert criticism is invited on the same evidential terms as outsider criticism.

 Ernos Labs is a separate revocable conformance designation requiring the public scientific constitution, transparent evidence, fail-closed engine rules and community standards. Submissions and counterexamples are invited through Maria.Smith.Sftoe@gmail.com and https://discord.gg/ucwGryVxGr.

34. Reproduction and invalidation

Run the repository-wide one-command verifier. A reviewer can invalidate a result by producing an omitted same-boundary coordinate, a second survivor, a nonminimal form, an induction counterexample, a failed inverse, an incomplete observation record, an uncorrected registered error mask or a mismatched receipt.

35. Scope and next branch

The next branch is Physics. It must test whether independently derived Fold relations correspond to mechanics, fields, spacetime, thermodynamics, quantum phenomena, gravitation, matter, waves, fluids, plasmas and condensed structures. Formal quantum computation cannot select the natural laws; sealed observation must do that work.

Foundation and full-field reconstruction roadmap — version 1.3.0

PUBLISHED SAME-PAPER SUCCESSOR VERSION 1.3.0. The preceding version 1.2.0 remains permanently preserved in the Zenodo version chain. This version publishes the complete-field Reversible and Quantum Computation roadmap while preserving all 22 admitted laws and their exact evidence boundaries.

All 22 published reversible and quantum-computation laws retain their receipts, including the separately forced unbounded-finite fault-tolerance result. Version 1.3 adds the complete operational, algorithmic, coding, verification, learning and limits roadmap without treating new fault models as automatic.

This amendment adds no scientific admission by prose. Every result already in the paper retains its exact receipt and evidence boundary. Every future result must pass the untouched engine independently. A branch described as complete to a dated inventory remains permanently open to lawful extension, correction, falsification and discoveries that satisfy the same public standard.

Scientific ownership

This branch owns reversible and quantum models of computation. It consumes Mathematics, Information Science and Classical Computation and supplies operational laws to Physics and later sciences without importing a conventional quantum formalism as a premise.

Layer One — foundational reconstruction

The foundational kernel derives:

1. reversible computation as a complete model;
2. one Fold distinction as an information unit;
3. exact state composition;
4. complete-word support as superposition-equivalent structure;
5. phase and interference operations;
6. entanglement as nonfactorable joint composition;
7. measurement and retained-record semantics;
8. reversible transformations and gates;
9. circuit syntax, semantics and resource accounting;
10. universality and classical-quantum correspondence;
11. quantum communication, coding and correction; and
12. limits of quantum computation.

Classical and quantum modes must be comparable branch by branch inside one exact Fold process, with no stochastic-collapse premise or complex proof scalar.

Layer Two — complete field reconstruction

The complete programme covers:

- reversible machines, reversible languages and thermodynamic record costs;
- information units, registers, state preparation and composition;

- superposition-equivalent supports and exact amplitude correspondence;
- phase, interference, path composition and observation;
- bipartite and multipartite entanglement structures;
- transformations, gate families and circuit semantics;
- quantum universality and compilation;
- algorithm families, speedup conditions and resource proofs;
- quantum complexity, reductions, completeness and lower bounds;
- communication, teleportation-equivalent transfer and distributed protocols;
- quantum coding, error detection and correction;
- exhaustive one-error and separately forced multi-error recovery;
- higher-fault models, thresholds and fault-tolerant composition;
- quantum simulation of finite physical systems;
- verification, interactive proof and delegated-computation structures;
- quantum cryptography and security interfaces;
- quantum learning and inference;
- operational classical-quantum equivalence and separation; and
- exact limits, including what finite support cannot establish about

unrestricted fault thresholds or computational advantage.

Publication sequence

The foundational paper establishes the Quantum Fold Machine. Later versions add complete operational, algorithmic, coding, fault-tolerance, verification and learning families. New fault models require new forced codes and cannot be treated as automatic extensions of the completed one-error construction.

Handoffs

Quantum physical measurements belong to Physics. Quantum chemistry and materials applications consume admitted quantum-computational laws but retain their own empirical evidence.

References

1. Landauer R. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*. 1961;5:183-191. doi:10.1147/rd.53.0183.
2. Bennett CH. Logical reversibility of computation. *IBM Journal of Research and Development*. 1973;17:525-532. doi:10.1147/rd.176.0525.
3. Feynman RP. Simulating physics with computers. *International Journal of Theoretical Physics*. 1982;21:467-488. doi:10.1007/BF02650179.
4. Deutsch D. Quantum theory, the Church-Turing principle and the universal quantum computer. *Proceedings of the Royal Society A*. 1985;400:97-117. doi:10.1098/rspa.1985.0070.
5. Shor PW. Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer. *SIAM Journal on Computing*. 1997;26:1484-1509. doi:10.1137/S0097539795293172.
6. Grover LK. A fast quantum mechanical algorithm for database search. *Proceedings of STOC*. 1996:212-219. doi:10.1145/237814.237866.
7. Schumacher B. Quantum coding. *Physical Review A*. 1995;51:2738-2747. doi:10.1103/PhysRevA.51.2738.
8. Smith M. Smithian Fold Theory Scientific Constitution. V3 clean-room repository, 2026.

9. Lundh A, Lexchin J, Mintzes B, Schroll JB, Bero L. Industry sponsorship and research outcome. *Intensive Care Medicine*. 2018. doi:10.1007/s00134-018-5293-7.

10. Fabbri A, Lai A, Grundy Q, Bero LA. The influence of industry sponsorship on the research agenda. *American Journal of Public Health*. 2018. PMID:30252531.

11. Gallo SA et al. Reliability and fairness in peer review of research funding. 2023. https://pmc.ncbi.nlm.nih.gov/articles/PMC10553257/.

12. Demicheli V, Di Pietrantonj C. Peer review for improving the quality of grant applications. *Cochrane Database of Systematic Reviews*. https://pmc.ncbi.nlm.nih.gov/articles/PMC8973940/.

13. Toward more published null and negative results. *Nature Communications*. 2025. https://pmc.ncbi.nlm.nih.gov/articles/PMC12459790/.

14. UNESCO. Recommendation on Open Science. 2021. https://www.unesco.org/en/legal-affairs/recommendation-open-science.